UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Homework 8: Array Corrections
Due Monday, October 19, 2026, 2:30 pm

Resources:
1. Chapter 6 of Howell
2. Appendix C of Howell

Hand in:
This assignment continues the correction of the data from HW6.

Remember, at all points, to explain what you do in your log and keep appropriate notes in your code.

1. (**5 points**) Use your terminal to make a new folder `hw8_<yourname>` under your `ast4762/homework` folder. For this homework your main homework file is a Python file named `hw8_<username>.py` (or a `hw8_<username>.ipynb`, but remember to save it as a `.py` file before submission). Save it in your homework folder. Remember to commit your files and push to GitHub. Start the file with the usual header. Then, include all your answers to Practicum 5.

2. (**20 points**) Write a function called `skycormednorm()` that implements sky subtraction using your normalized sky frame.

The function should accept two 2D data arrays:
- the object frame and
- the normalized sky frame
  (both should be already dark-subtracted).

The function should be able to calculate the normalization factor by using an *optional keyword* for the normalization region, similar to function `normmedcomb()`. The default region should be the whole array.

The function should then de-normalize the sky, subtract it from the object frame, and return the resulting sky-subtracted frame. Be careful not to modify any data in the caller.

Apply `skycormednorm()` in a loop to all the object data.

Write the last resulting frame to a FITS file named `stars_13s_9_nosky.fits`.

Print the value of pixel [217, 184] of this last frame.

3. (**5 points**) Test that your code works on Stokes and submit your homework.